

Problem

A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h. How much charge can it release at that rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?

Step-by-step solution

Step 1 of 2

Consider the following data:

The current delivered by a rechargeable flashlight battery, $i = 90 \text{ mA}$

The time that the current delivered, 12 h .

Calculate the charge that can be release.

$$\begin{aligned} q &= it \\ &= (90 \times 10^{-3})(12 \text{ h} \times 60 \text{ min/h} \times 60 \text{ s/min}) \\ &= 3.888 \text{ kC} \end{aligned}$$

Therefore, the charge that can be release is $\boxed{3.888 \text{ kC}}$.

Step 2 of 2

The terminal voltage of a rechargeable flashlight battery, $v = 1.5 \text{ V}$

Calculate the power delivered by the rechargeable flashlight battery.

$$\begin{aligned} p &= vi \\ &= (1.5)(90 \times 10^{-3}) \\ &= 135 \text{ mW} \end{aligned}$$

Calculate the energy delivered by the battery.

$$\begin{aligned} w &= pt \\ &= (135 \times 10^{-3})(12 \text{ h} \times 60 \text{ min/h} \times 60 \text{ s/min}) \\ &= 5.832 \text{ kJ} \end{aligned}$$

Therefore, the energy delivered by the battery is $\boxed{5.832 \text{ kJ}}$.